

7.5 Determine the z-transform and its ROC for

$$x[n] = 2\left(\frac{2}{3}\right)^n u[n] - \left(\frac{2}{5}\right)^n u[n]$$

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資訊116

$$x_1[n] = 2\left(\frac{2}{3}\right)^n u[n]$$

$$X_1(z) = \frac{2z}{z - \frac{2}{3}}, \quad |z| > \frac{2}{3}$$

$$x_2[n] = -\left(\frac{2}{5}\right)^n u[n]$$

$$X_2(z) = \frac{-z}{z - \frac{2}{5}}, \quad |z| > \frac{2}{5}$$

$$X(z) = \frac{2z}{z - \frac{2}{3}} - \frac{z}{z - \frac{2}{5}} = \frac{6z}{3z - 2} - \frac{5z}{5z - 2}$$

$$\text{ROC: } |z| > \frac{2}{3} \neq$$

7.6 Find the z-transform of the following signals:

(a) $u[n - m]$

(b) $na^n u[n]$

(c) $a^n \cos \pi n u[n]$

$$(a) \mathcal{Z}\{u[n-m]\}$$

$$= z^{-m} \frac{z}{z-1}$$

$$= \frac{z^{-m+1}}{z-1} \quad \#$$

$$(b) \mathcal{Z}\{na^n u[n]\}$$

$$= -z \frac{d}{dz} \frac{z}{z-a}$$

$$= -z \frac{(z-a) - z}{(z-a)^2}$$

$$= \frac{az}{(z-a)^2} \quad \#$$

$$(c) \mathcal{Z}\{a^n \cos \pi n u[n]\}$$

$$= \mathcal{Z}\{a^n (-1)^n u[n]\}$$

$$= \mathcal{Z}\{(-a)^n u[n]\}$$

$$= \frac{z}{z+a} \quad \#$$

7.20 The z-transform of a discrete-time signal is

$$X(z) = \frac{2z}{z^2 + 3z + 1}$$

Find the z-transform of the following signals:

(a) $y[n] = x[n-1]u[n-1]$

(b) $y[n] = \sin(\pi n/4) x[n]$

(c) $y[n] = n^2 x[n]$

(d) $y[n] = 2x[n]^* x[n]$

$$(a) Y(z) = z^{-1} X(z) = \frac{z}{z^2 + 3z + 1}$$

$$(b) Y(z) = \frac{j}{2} (X(e^{j\frac{\pi}{4}} z) - X(e^{-j\frac{\pi}{4}} z))$$

$$= \frac{j}{2} \left(\frac{ze^{j\frac{\pi}{4}}}{(e^{j\frac{\pi}{4}} z)^2 + 3(e^{j\frac{\pi}{4}} z) + 1} - \frac{ze^{-j\frac{\pi}{4}}}{(e^{-j\frac{\pi}{4}} z)^2 + 3(e^{-j\frac{\pi}{4}} z) + 1} \right) *$$

$$(c) Y(z) = -z \frac{d}{dz} \left(-z \frac{d}{dz} X(z) \right)$$

$$= -z \frac{d}{dz} \frac{2z^3 - 2z}{(z^2 + 3z + 1)^2}$$

$$= \frac{2z^5 - 6z^4 - 12z^3 - 6z^2 + 2z}{(z^2 + 3z + 1)^3} *$$

$$\frac{d}{dz} \frac{2z}{z^2 + 3z + 1}$$

$$= \frac{2(z^2 + 3z + 1) - 2z(2z + 3)}{(z^2 + 3z + 1)^2}$$

$$= \frac{-2z^2 + 2}{(z^2 + 3z + 1)^2}$$

$$(d) Y(z) = 2X(z)^2$$

$$= \frac{8z^2}{(z^2 + 3z + 1)^2} *$$

$$\frac{d}{dz} \frac{-2z^2 + 2}{(z^2 + 3z + 1)^2}$$

$$= \frac{(6z^2 + 2)(z^2 + 3z + 1)^2 - (-2z^2 + 2) \cdot 2(z^2 + 3z + 1)(2z + 3)}{(z^2 + 3z + 1)^4}$$

$$= \frac{2z^4 - 6z^3 - 12z^2 - 6z + 2}{(z^2 + 3z + 1)^3} \quad \frac{-6z^4 + 18z^3 + 6z^2 + 2 - 8z^4 + 12z^3 + 6z^2 + 2}{2z^4 - 6z^3 - 12z^2 - 6z + 2}$$

7.21 Using the z-transform, determine the convolution of these sequences:

$$x[n] = [1, -1, 3, 2], \quad h[n] = [1, 0, 2, 1, -3].$$

$$X(z) = 1 - z^{-1} + 3z^{-2} + 2z^{-3}$$

$$H(z) = 1 + 2z^{-2} + z^{-3} - 3z^{-4}$$

$$Y(z) = X(z)H(z)$$

$$= 1 + 2z^{-2} + z^{-3} - 3z^{-4}$$

$$- z^{-1} - 2z^{-3} - z^{-4} + 3z^{-5}$$

$$+ 3z^{-2} + 6z^{-4} + 3z^{-5} - 9z^{-6}$$

$$+ 2z^{-3} + 4z^{-5} + 2z^{-6} - 6z^{-7}$$

$$= 1 - z^{-1} + 5z^{-2} + z^{-3} + 2z^{-4} + 10z^{-5} - 7z^{-6} - 6z^{-7}$$

$$y[n] = [1, -1, 5, 1, 2, 10, -7, -6]$$

7.24 Find the inverse z-transform of the following functions:

(a) $X(z) = \frac{2z}{z^2 - z + 1}$

(b) $Y(z) = \frac{z(z+2)}{(z+1)(z-1)}$

(c) $H(z) = \frac{4z}{(z^2 + z + 1)(z + 1/2)}$

$$\begin{aligned} (a) \mathcal{Z}\{\sin \Omega n u[n]\} &= \frac{j}{2} \left(\frac{e^{j\Omega} z}{e^{j\Omega} z - 1} - \frac{e^{-j\Omega} z}{e^{-j\Omega} z - 1} \right) \\ &= \frac{j}{2} \frac{z(z - e^{j\Omega} - z + e^{-j\Omega})}{z^2 - 2 \cos \Omega z + 1} \\ &= \frac{\sin \Omega z}{z^2 - 2 \cos \Omega z + 1} \end{aligned}$$

$$\begin{aligned} X(z) &= \frac{2z}{z^2 - z + 1}, \cos \Omega = \frac{1}{2}, \Omega = \frac{\pi}{3} \\ &= \frac{\frac{4}{\sqrt{3}} \sin \frac{\pi}{3} z}{z^2 - 2 \cos \frac{\pi}{3} z + 1} \end{aligned}$$

$$x[n] = \frac{4}{\sqrt{3}} \sin\left(\frac{\pi}{3} n\right) u[n] \quad \#$$

$$(b) \frac{Y(z)}{z} = \frac{z+2}{(z+1)(z-1)}$$

$$= \frac{-0.5}{z+1} + \frac{1.5}{z-1}$$

$$Y(z) = \frac{-0.5z}{z+1} + \frac{1.5z}{z-1}$$

$$y[n] = (-0.5(-1)^n + 1.5) u[n] \quad \#$$

$$\begin{aligned} (c) \frac{H(z)}{z} &= \frac{4}{(z^2 + z + 1)(z + \frac{1}{2})} \\ &= \frac{Az+B}{z^2 + z + 1} + \frac{C}{z + \frac{1}{2}} \quad C = \frac{4}{\frac{1}{4} - \frac{1}{2} + 1} \\ &= \frac{-\frac{16}{3}z - \frac{8}{3}}{z^2 + z + 1} + \frac{\frac{16}{3}}{z + \frac{1}{2}} \quad = \frac{\frac{16}{3}}{\frac{1}{3} + \frac{1}{2}B - 4} \end{aligned}$$

$$\begin{aligned} H(z) &= \frac{16}{3} \left(\frac{z}{z + 0.5} - \frac{z(z + 0.5)}{z^2 + z + 1} \right) \\ \cos \Omega &= -\frac{1}{2}, \Omega = \frac{2\pi}{3} \end{aligned}$$

$$H(z) = \frac{16}{3} \left(\frac{z}{z + 0.5} - \frac{z(z - \cos \frac{2\pi}{3})}{z^2 - 2 \cos \frac{2\pi}{3} z + 1} \right)$$

$$h[n] = \frac{16}{3} \left((-0.5)^n - \cos\left(\frac{2\pi}{3} n\right) u[n] \right) \quad \#$$

7.27 Invert each of the following z-transform:

$$(a) X_1(z) = \frac{1 - z^{-1}}{1 - z^{-1} - 0.75z^{-2}}$$

$$(b) X_2(z) = \frac{1 + z^{-1}}{1 - 0.8z^{-1} + 0.64z^{-2}}$$

$$\begin{aligned} (a) X_1(z) &= \frac{1 - z^{-1}}{1 - z^{-1} - 0.75z^{-2}} \\ &= \frac{A}{(1 - 1.5z^{-1})(1 + 0.5z^{-1})} \quad \frac{\frac{1}{3}}{\frac{-4}{3}} = 0.25 \\ &= \frac{0.25}{1 - 1.5z^{-1}} + \frac{0.15}{(1 + 0.5z^{-1})} \end{aligned}$$

$$x_1[n] = (0.25(1.5)^n + 0.15(-0.5)^n) u[n] \quad \#$$

$$\begin{aligned} (b) X_2(z) &= \frac{1 + z^{-1}}{1 - 0.8z^{-1} + 0.64z^{-2}} \\ &= \frac{z^2 + z}{z^2 - 0.8z + 0.64} \\ &= \frac{(\frac{z}{0.8})^2 + \frac{z}{0.8}}{(\frac{z}{0.8})^2 - \frac{z}{0.8} + 1} \\ &= \frac{(\frac{z}{0.8})^2 - \frac{z}{0.8} + 1}{(\frac{z}{0.8})^2 - \frac{z}{0.8} + 1} + \frac{1.75 \frac{z}{0.8}}{(\frac{z}{0.8})^2 - \frac{z}{0.8} + 1} \\ &= \frac{\frac{z}{0.8} (\frac{z}{0.8} - \cos \frac{\pi}{3}) + \frac{3.5}{\sqrt{3}} \sin \frac{\pi}{3} \frac{z}{0.8}}{(\frac{z}{0.8})^2 - 2 \cos \frac{\pi}{3} \frac{z}{0.8} + 1} \quad 1.75 \times \frac{z}{\sqrt{3}} = \frac{3.5}{\sqrt{3}} \end{aligned}$$

$$x_2[n] = (0.8)^n \left(\cos \frac{\pi}{3} n + \frac{1}{2\sqrt{3}} \sin \frac{\pi}{3} n \right) u[n] \quad \#$$

7.31 Using the z-transform, solve the following difference equation:

$$y[n+1] - 2y[n] = (1.5)^n, \quad y[0] = 1.$$

$$zY(z) - zy[0] - 2Y(z) = \frac{z}{z-1.5}$$

$$\Rightarrow (z-2)Y(z) = \frac{z}{z-1.5} + z$$

$$\Rightarrow (z-2)Y(z) = \frac{z^2 - 0.5z}{z-1.5}$$

$$\Rightarrow Y(z) = \frac{z(z-0.5)}{(z-2)(z-1.5)}$$

$$\Rightarrow \frac{Y(z)}{z} = \frac{z-0.5}{(z-2)(z-1.5)}$$

$$= \frac{3}{z-2} - \frac{1}{z-1.5}$$

$$Y(z) = \frac{3z}{z-2} - \frac{2z}{z-1.5}$$

$$y[n] = (3 \cdot 2^n - 2 \cdot (1.5)^n) u[n]$$

7.36 The transfer function of a discrete-time system is

$$H(z) = \frac{1 + 2z^{-1}}{1 - z^{-1} + z^{-2}}$$

Find the system response $y[n]$ when the input is a unit step function $u[n]$.

$$Y(z) = \frac{z(z^2 + 2z)}{(z-1)(z^2 - z + 1)}$$

$$\frac{Y(z)}{z} = \frac{z(z+2)}{(z-1)(z^2 - z + 1)}$$

$$= \frac{z}{z-1} + \frac{-2z+3}{z^2 - z + 1}$$

$$Y(z) = \frac{z}{z-1} + \frac{-2(z^2 - 1.5z)}{z^2 - z + 1}$$

$$= \frac{z}{z-1} + \frac{-2(z(z - \cos \frac{\pi}{3}) - \frac{2}{\sqrt{3}} \sin \frac{\pi}{3} z)}{z^2 - 2 \cos \frac{\pi}{3} z + 1}$$

$$y[n] = \left[3 - 2 \left(\cos \left(\frac{\pi}{3} n \right) - \frac{2}{\sqrt{3}} \sin \left(\frac{\pi}{3} n \right) \right) \right] u[n] *$$

7.43 Obtain the impulse and step responses of the discrete-time system with transfer function

$$H(z) = \frac{0.8z}{(z-0.6)(z-2)}$$

impulse: $\frac{Y(z)}{z} = \frac{0.8}{(z-0.6)(z-2)}$

$$= \frac{-\frac{4}{7}}{z-0.6} + \frac{\frac{4}{7}}{z-2}$$

$$Y(z) = \frac{-\frac{4}{7}z}{z-0.6} + \frac{\frac{4}{7}z}{z-2}$$

$$y[n] = \frac{4}{7} (2^n - (0.6)^n) u[n]$$

step: $Y(z) = \frac{z}{z-1} \cdot \frac{0.8z}{(z-0.6)(z-2)}$

$$\frac{Y(z)}{z} = \frac{-2}{z-1} + \frac{\frac{6}{7}}{z-0.6} + \frac{\frac{8}{7}}{z-2}$$

$$y[n] = \left[-2 + \frac{6}{7} (0.6)^n + \frac{8}{7} \cdot 2^n \right] u[n] \quad *$$

B. Problem X

For a signal recording system with a sampling rate of 500 Hz, design a digital filter by placing the zeros at 60 Hz and 120 Hz in the z-plane.

$$60 \text{ Hz: } \omega_1 = \pm \frac{60}{500} 2\pi = \pm 0.7536 = \pm 43.2^\circ$$

$$120 \text{ Hz: } \omega_2 = \pm \frac{120}{500} 2\pi = \pm 1.5072 = \pm 86.4^\circ$$

$$z_{1,2} = \cos \omega_1 \pm j \sin \omega_1 = 0.7290 \pm j 0.6845$$

$$z_{3,4} = \cos \omega_2 \pm j \sin \omega_2 = 0.0628 \pm j 0.9980$$

$$\begin{aligned} H_{60}(z) &= (1 - z^{-1} z_1)(1 - z^{-1} z_2) \\ &= 1 - (z_1 + z_2) z^{-1} + (z_1 z_2) z^{-2} \\ &= 1 - 1.4580 z^{-1} + z^{-2} \end{aligned}$$

$$H_{120}(z) = 1 - 0.1256 z^{-1} + z^{-2}$$

$$\begin{aligned} H(z) &= (1 - 1.4580 z^{-1} + z^{-2})(1 - 0.1256 z^{-1} + z^{-2}) \\ &= 1 - 1.5836 z^{-1} + 2.1831 z^{-2} - 1.5836 z^{-3} + z^{-4} \end{aligned}$$

$$H(z=1) = 1 - 1.5836 + 2.1831 - 1.5836 = 1.0159$$

$$H'(z) = \frac{1}{1.0159} (1 - 1.5836 z^{-1} + 2.1831 z^{-2} - 1.5836 z^{-3} + z^{-4})^*$$

7.45 Use MATLAB to find the inverse z-transform of

$$X(z) = \frac{z}{z - 0.6}$$

```
>> HW4_7_45
```

```
X =
```

```
z/(z - 3/5)
```

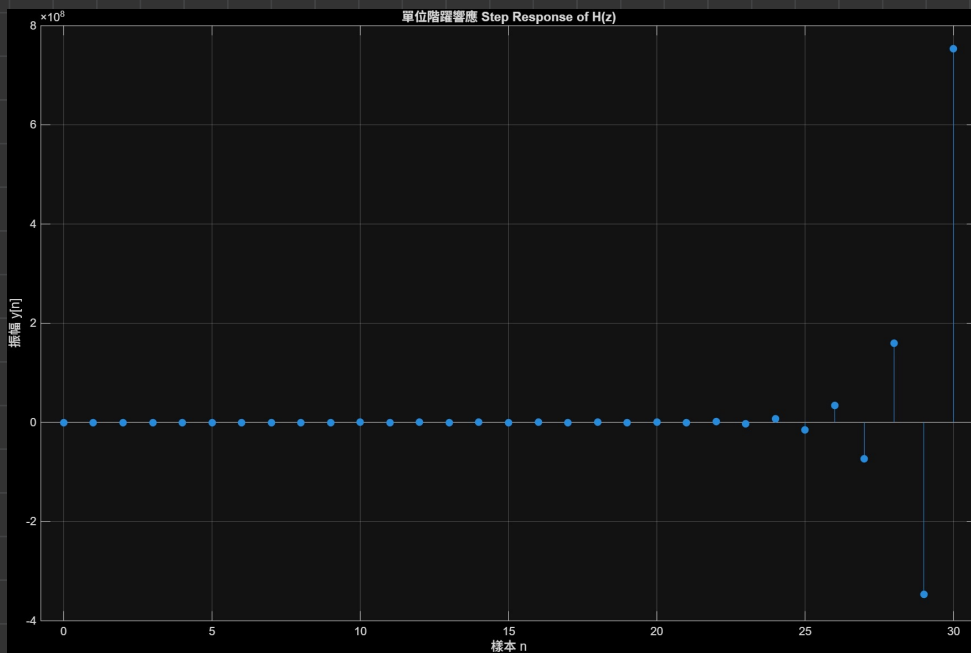
```
x =
```

```
(3/5)^n
```

7.46 A linear discrete-time system is represented by the transfer function

$$H(z) = \frac{z+1}{z^3 + 2z^2 + z + 3}$$

Use MATLAB to plot the step response of the system.



7.47 Determine the poles and zeros of the transfer function

$$H(z) = \frac{z^2 + 6z + 1}{z^4 + 3z^3 + 4z + 10}$$

```
>> HW4_7_47
```

```
零點 (Zeros):
```

```
-5.8284
```

```
-0.1716
```

```
極點 (Poles):
```

```
-3.0794 + 0.0000i
```

```
0.7401 + 1.3305i
```

```
0.7401 - 1.3305i
```

```
-1.4009 + 0.0000i
```

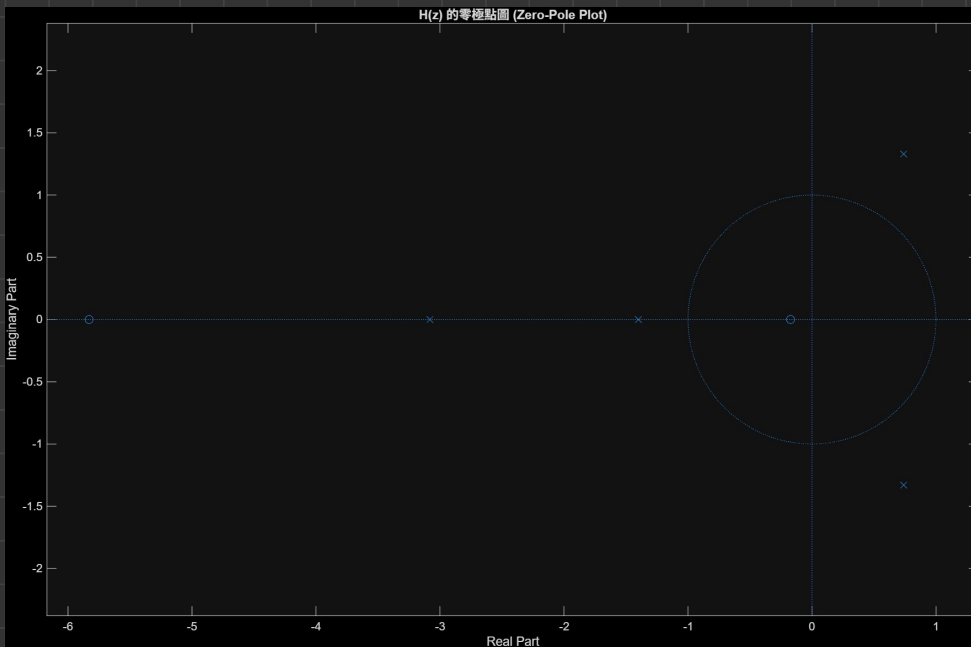
```
極點模值 (Magnitude of Poles):
```

```
3.0794
```

```
1.5225
```

```
1.5225
```

```
1.4009
```



7.50 Check the stability of a system described by the following transfer function:

$$H(z) = \frac{z^3 - 2z^2 + 6z + 1}{z^5 - 2z^4 + 5z^2 - z + 4}$$

```
>> HW4_7_50
```

系統極點為：

```
1.6969 + 1.1485i
```

```
1.6969 - 1.1485i
```

```
-1.4722 + 0.0000i
```

```
0.0391 + 0.8035i
```

```
0.0391 - 0.8035i
```

極點模值 (Magnitude) 為：

```
2.0491
```

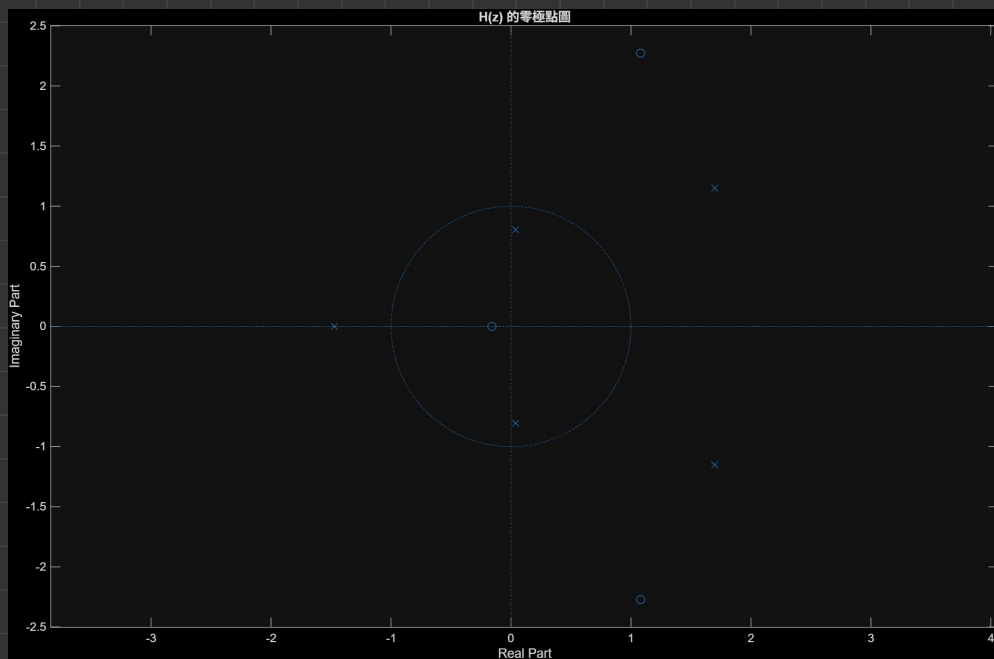
```
2.0491
```

```
1.4722
```

```
0.8044
```

```
0.8044
```

結論：有極點位於單位圓外或圓上，系統不穩定 (Unstable)。



B. Plot the magnitude response of the filter you design in Problem X in linear and logarithmic (dB) scales using 1024-pt fft,

